

> PROTOCOL_OVERVIEW

- DASHBOARD >
- VAULT STATUS
- STRATEGIES
- MARKETS
- RISK ENGINE
- PERFORMANCE
- CONFIG

> SYSTEM LOG

[12:00:01] SYSTEM BOOT OK
 [12:00:02] AI ENGINE OK
 [12:00:03] DATA FEED OK
 [12:00:04] RISK ENGINE OK
 [12:00:05] VAULT SYSTEM OK
 [12:00:06] ARBITRAGE MODULE OK

> ALL SYSTEMS OPERATIONAL

POLYALPHA PROTOCOL

AI x DeFi x Prediction Markets

An autonomous AI-driven vault for decentralized prediction market arbitrage.



[TESTNET LIVE]



[PAPER TRADING ACTIVE]

TVL



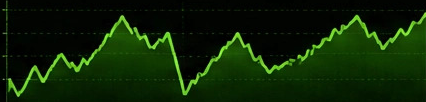
\$1,247,389.42



DAILY PNL



+\$18,732.64
 (+1.52%)



WIN RATE



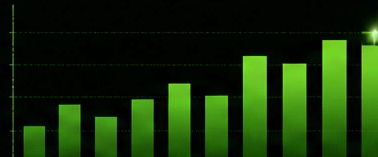
62.9%



SHARPE RATIO



2.53



ChainLab Testnet



(chainId: 31337)



May 14, 2026



William Yong



AGENDA

- 01 Problem Definition
- 02 Market Opportunity
- 03 Solution Overview
- 04 How It Works
- 05 Alpha Engine Mechanism
- 06 Live Demonstration
- 07 Technical Architecture
- 08 Security Model
- 09 Tokenomics
- 10 Open-Source Integration
- 11 Competitive Landscape
- 12 Strategy Validation
- 13 Business Model
- 14 Roadmap
- 15 Conclusion & Q&A





PROBLEM DEFINITION

Prediction markets are broken for retail - speed, data, and capital create an impossible barrier.



NO SPEED

Retail users cannot react in milliseconds.



NO DATA

Real-time CLOB data requires expensive infrastructure.



NO CAPITAL

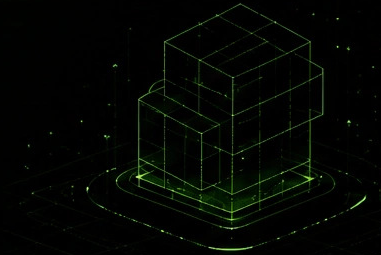
Minimum viable arbitrage requires \$10,000+.



01

\$63.5B

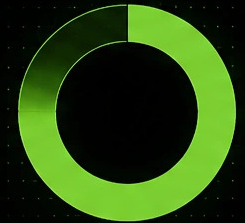
Polymarket 2025 Trading Volume



02

97.5%

Market share: Polymarket + Kalshi



03

0

Institutional-grade retail tools available.





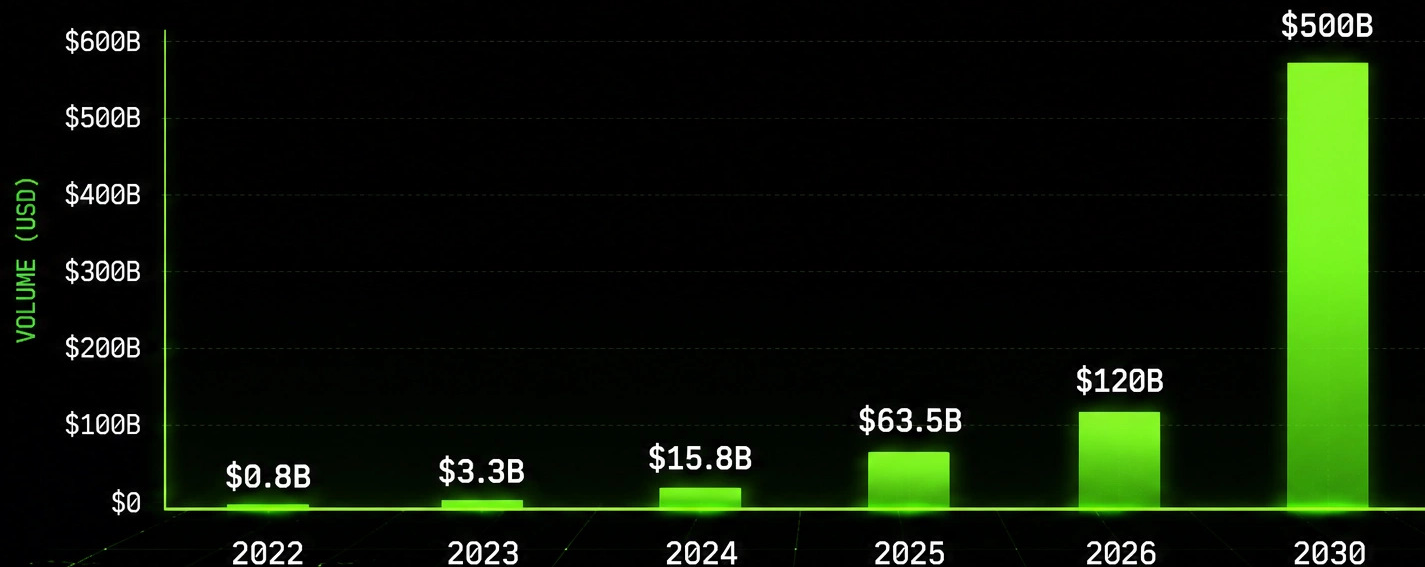
> MARKET OPPORTUNITY

PREDICTION MARKETS GREW 330% CAGR – THE ARBITRAGE INFRASTRUCTURE LAYER IS STILL EMPTY.



GLOBAL PREDICTION MARKET VOLUME

Historical & Projected (USD)



Source: Dune Analytics, Messari, Precedence Research, Custom Market Model



MARKET TIERS





[SOLUTION OVERVIEW]

SOLUTION OVERVIEW

PolyAlpha democratizes institutional arbitrage through three interlocking pillars.

01

ERC-4626 VAULT



On-chain asset management,
100% non-custodial.

02

AI ALPHA ENGINE



Off-chain signal generation
via B.ai: GLM-5.1, Claude 4.6,
GPT-5.5.

03

DAO GOVERNANCE



Community-owned protocol,
buyback-and-burn.



62.9%

WIN RATE



2.53

SHARPE



\$10

MINIMUM DEPOSIT

From deposit to yield in 6 automated steps - no human intervention required.

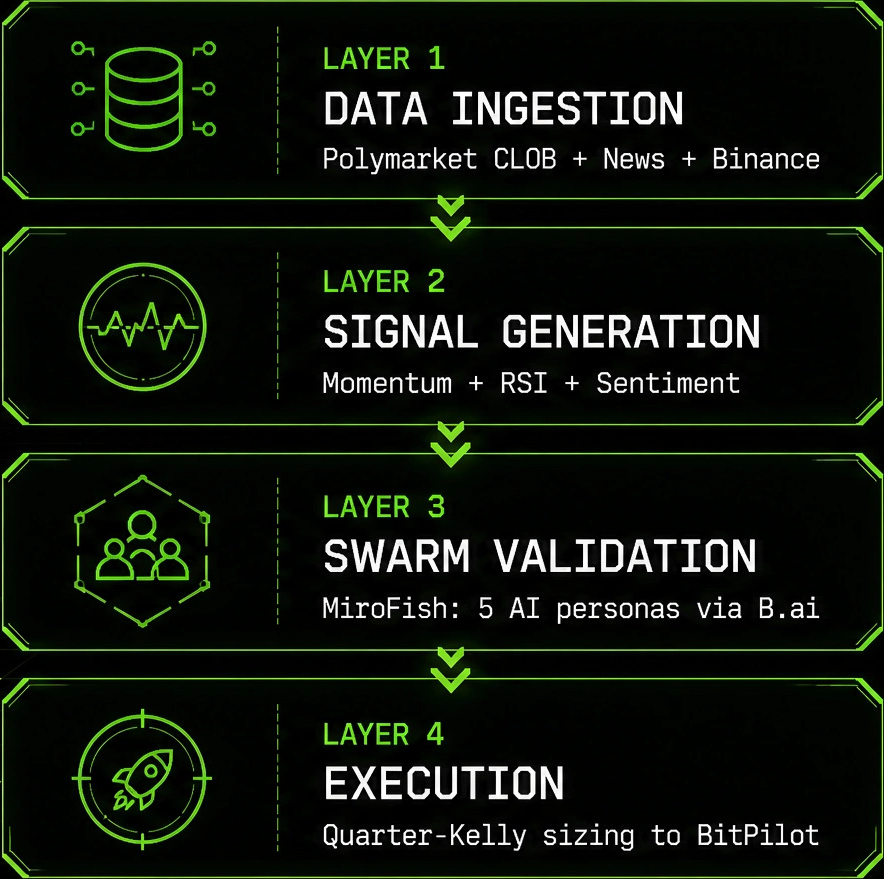


FEATURE	 POLYALPHA PROTOCOL	TRADITIONAL HEDGE FUND
 Min Deposit	\$10 USDC	\$1,000,000+
 Transparency	100% On-Chain	Black Box
 Fee Structure	0.5/20 no lock-up	2/20 + 2-year lock
 Circuit Breaker	20% hardcoded	Discretionary



Alpha Engine Mechanism

A 4-layer signal stack ensures we only trade when the mathematical edge is confirmed.





Live Demonstration

Every trade, every signal, every dollar - visible on-chain in real-time.



VAULT STATUS



TVL

\$47,230 USDC



Shares

47,230 PALPHA



APY

18.7%



AI SIGNALS



STRONG_BUY

(BTC/USDT momentum)



Confidence

87%



Swarm

4/5 consensus



TRADE HISTORY

TIME (UTC)	PAIR	SIDE	SIZE (USDC)	PNL (USDC)
2025-05-26 14:32:18	BTC/USDT	BUY	10,000	+842.16
2025-05-26 14:18:07	ETH/USDT	BUY	7,500	+512.47
2025-05-26 14:05:55	SOL/USDT	SELL	5,000	-213.34
2025-05-26 13:52:31	BTC/USDT	BUY	8,000	+681.23
2025-05-26 13:41:09	ETH/USDT	BUY	6,000	+378.90

Showing 5 of 128 recent trades

Total PNL (Today): +2,201.42 USDC



SWARM PANEL



DAVE

BUY



TINA

BUY



MIKE

BUY



RITA

BUY



CARL

BUY

SWARM
CONSENSUS



60%

BUY

3 OUT OF 5 AGENTS RECOMMEND BUY



LIVE: polyalpha-dashboard.vercel.app



Open now on your phone.



Technical Architecture

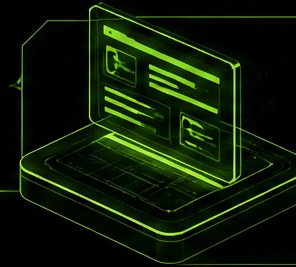
6 smart contracts, 3 system layers — fully modular and auditable.

SMART CONTRACTS (Solidity 0.8.25 • ChainLab Testnet)

#	SMART CONTRACT	PURPOSE & STANDARD
01	 PolyAlphaVault.sol	Asset management, ERC-4626
02	 PALPHAToken.sol	Governance token, ERC-20
03	 PALPHABuybackBurn.sol	Deflationary burn, Custom
04	 PALPHAGovernance.sol	DAO voting, OpenZeppelin
05	 PALPHASTaking.sol	Yield distribution, Custom
06	 PALPHAOracle.sol	Price feeds, Chainlink-compatible

3 SYSTEM LAYERS

1



FRONTEND LAYER

React + TailwindCSS + Vercel

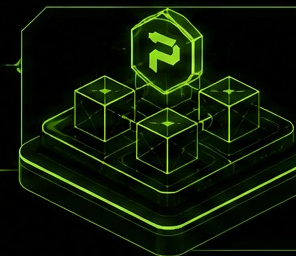
2



OFF-CHAIN AGENT LAYER

Python + GLM-5.1 + MiroFish + B.ai

3



ON-CHAIN LAYER

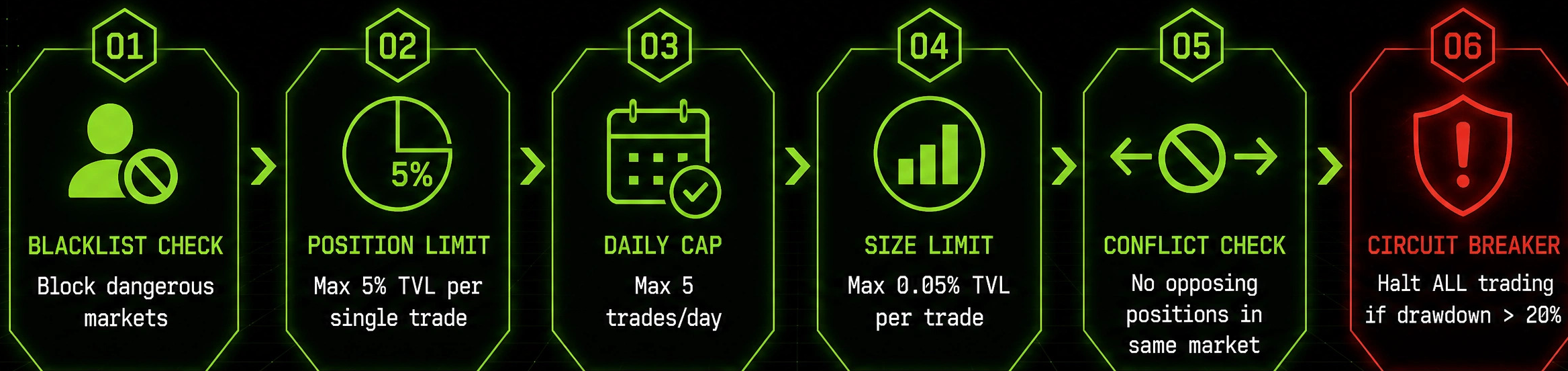
ChainLab Testnet • OpenZeppelin v5



SECURITY MODEL

6-layer BitPilot Safety Chain –

hardcoded in smart contract, impossible to override.



WARNING: CIRCUIT BREAKER: max_drawdown_bps = 2000 (20%) is hardcoded in **PolyAlphaVault.sol** – no admin override possible

PALPHA: a deflationary governance token aligned with protocol performance.

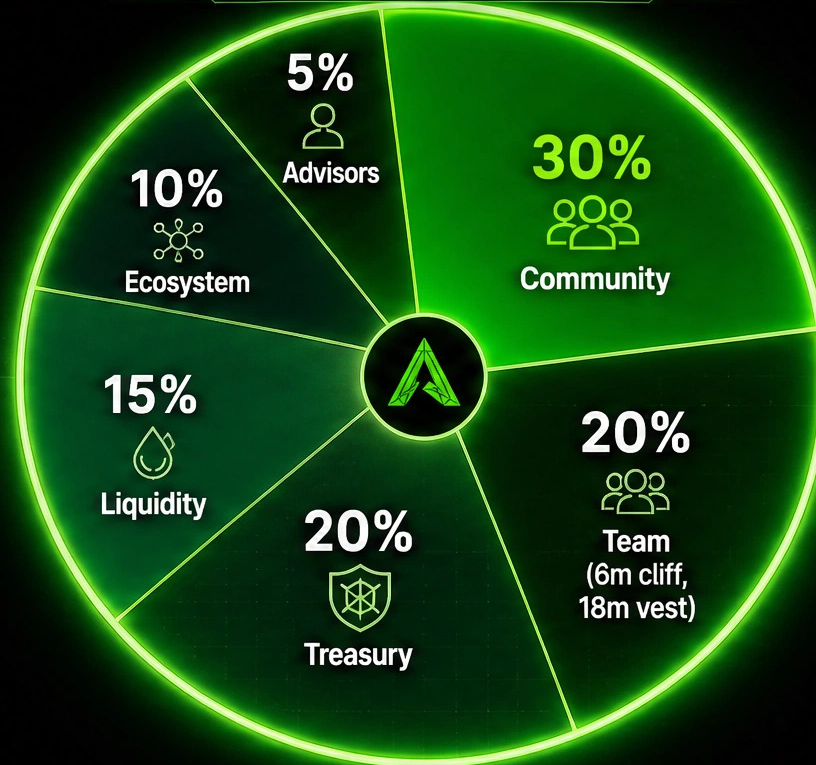
PALPHA FLYWHEEL



MAIN STATS

TOTAL SUPPLY

10,000,000
PALPHA



BURN MECHANISM



30%
of revenue used for
buyback-and-burn

STAKING APY



10%
(Fixed in v1)

Standing on the shoulders of giants — 6 battle-tested open-source protocols.

1



MiroFish

666ghj/MiroFish

- Multi-Agent Swarm AI
- 5 personas
- GLM-5.1 + Zep memory

2



polymarket-toolkit

runesleo

- CLOB API + PnL Engine
- Real-time order book

3



daily-news

6551Team

- News Sentiment API
- Bullish/Bearish scoring

4



bgtask

duolaAmengweb3

- BitPilot Safety Chain
- 6-step safety gate

5



Microsoft Qlib

- Backtesting Framework
- Alpha158 factor library
- Sharpe 2.53 verified

6



CAMEL-OASIS

camel-ai

- Agent Memory Framework
- Zep Cloud memory



All integrations are **REAL** code — not mock functions. Verified in **agent/** directory.

Competitive Landscape.

PolyAlpha is the only protocol combining AI + DeFi vault + prediction market arbitrage.

DIMENSION	 POLYALPHA	POLYMARKET	AUGUR	NUMERAI	YEARN FINANCE
 Min Deposit	\$10 USDC	N/A	N/A	\$1	\$100+
 AI Strategy	4-Layer + Swarm	No	No	ML Models	No
 On-Chain Transparency	100%	Partial	Partial	Partial	100%
 Prediction Market Focus	YES	YES	YES	Indirect	No
 Automated Execution	YES	No	No	No	YES
 Governance Token	PALPHA	No	Reputation	NMR	YFI
 Fee Structure	0.5/20 aligned	Up to 2%	1-2%	10%	0-2%



KEY INSIGHT: Only PolyAlpha combines automated AI arbitrage + non-custodial vault + prediction market focus + aligned tokenomics in a single protocol.

STRATEGY VALIDATION

Backtested on Polymarket data: **Sharpe 2.53, Win Rate 62.9%.**

CUMULATIVE PNL - JAN 2024 TO MAY 2026



WIN RATE

62.9%



SHARPE RATIO

2.53



MAX DRAWDOWN

-13.09%



TOTAL TRADES

97
trades



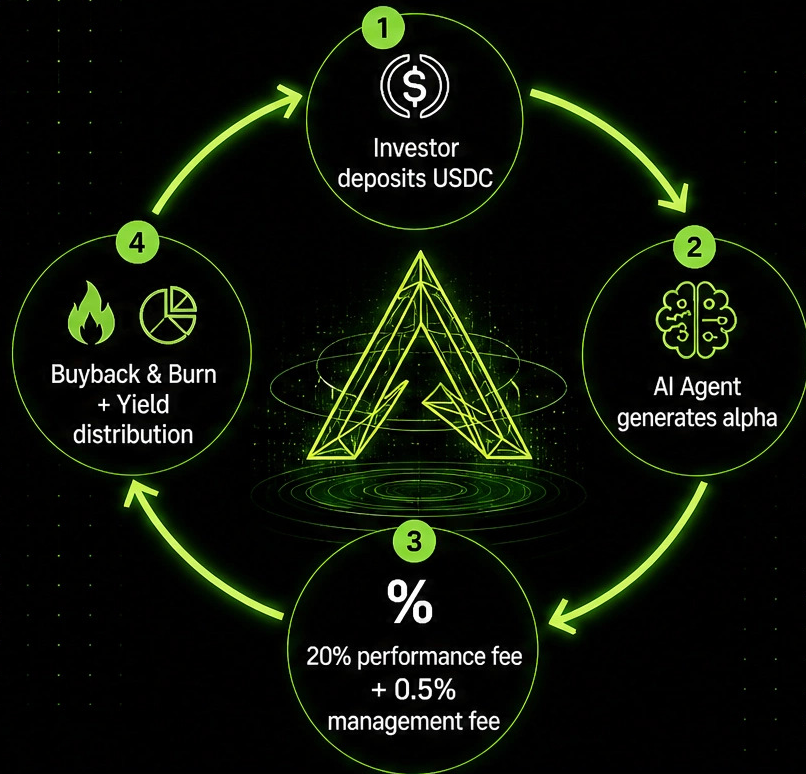
BACKTEST PERIOD

SPECIFIC
REGIME

Business Model

0.5/20 hedge fund structure - aligned incentives, sustainable revenue from day one.

REVENUE CYCLE



0.5% / 20%

Management + Performance

REVENUE PROJECTION

1



\$1M TVL:

\$42,400 / year

2



\$5M TVL:

\$212,000 / year

3



\$10M TVL

(Phase 3 Target):

\$424,000 / year



Fee structure mirrors top hedge funds but with full on-chain transparency and no lock-up period.

Roadmap & Future Vision

[Three phases from testnet to \$10M TVL institutional protocol.]



PHASE 1 FOUNDATION (COMPLETE)

- ✓ 6 smart contracts deployed on ChainLab
- ✓ ERC-4626 Vault live
- ✓ Python AI Agent running
- ✓ React Dashboard on Vercel
- ✓ 6 open-source integrations

100%



PHASE 2 EDGE BUILD (IN PROGRESS)

- ✓ MiroFish Swarm AI integrated
- ✓ EdgeBuild UI deployed
- Mainnet deployment
- First real trades
- Community beta testing

60%



PHASE 3 SCALE (Q4 2026)

- \$10M TVL target
- DAO governance live
- Multi-chain expansion
- Institutional partnerships
- PALPHA token launch

0%

Conclusion & Call to Action

[PolyAlpha: where institutional alpha meets DeFi transparency.]



6
**SMART CONTRACTS
DEPLOYED**
(ChainLab Testnet)



62.9%
**BACKTEST
WIN RATE**
(Sharpe Ratio 2.53)



6
**OPEN-SOURCE
INTEGRATIONS**
(Real code, not mock)



\$424K
**PROJECTED
ANNUAL REVENUE**
(at \$10M TVL)

>_ TERMINAL

> polyalpha-protocol --status

☑ PHASE 1: COMPLETE [6/6 contracts deployed]

⦿ PHASE 2: IN PROGRESS [MiroFish + EdgeBuild live]

→ LIVE DEMO: polyalpha-dashboard.vercel.app

→ SOURCE CODE: github.com/YongWilliam-ai/polyalpha-protocol



[Thank you. Questions welcome.]